Apache Pig — Detailed Explanation

Apache Pig is a high-level data processing platform that is mainly used to process and analyze large amounts of data in the Hadoop ecosystem.

To understand why Pig is needed, imagine a company has millions or billions of records. For example, an online shopping company may have data about customers, products, orders, payments, and website activity. Processing such a huge amount of data using normal programming can be difficult. Hadoop provides a way to process this large data, but traditionally we may need to write MapReduce programs, which can be lengthy and complicated. Apache Pig makes this process easier.

Instead of writing a complex MapReduce program, we can write simple instructions using a language called **Pig Latin**.

What is Pig Latin?

Pig Latin is the scripting language used by Apache Pig. It contains simple commands that describe what we want to do with the data.

LOAD → load data
FILTER → select required records
GROUP → group records
JOIN → combine datasets
ORDER → sort data
STORE → save the result

Example: Student Data

Suppose a college has a huge student dataset:

```
Name,Branch,Marks
Asha,CSE,85
Ravi,ISE,55
Manu,CSE,75
Priya,AI&ML,90
Kiran,ISE,40
```

Suppose the college has millions of student records and we want to find students who scored more than 60. In Pig, we can write:

```
students = LOAD 'students.csv'
USING PigStorage(',')
AS (name:chararray, branch:chararray, marks:int);

passed = FILTER students BY marks > 60;

STORE passed INTO 'output';
```

Step 1 — LOAD

LOAD reads the data from the file. Apache Pig takes the student data and loads it for processing.

Step 2 — FILTER

FILTER selects records based on a condition. Here, Pig selects students whose marks are greater than 60.

```
Asha     CSE      85
Manu     CSE      75
Priya    AI&ML    90
```

Step 3 — STORE

STORE saves the processed result in the specified output location.

Complete process: Input Data → LOAD → FILTER → STORE → Output

How Apache Pig Works Internally

When we write Pig Latin, Pig does not simply execute each command directly. First, Pig takes the Pig Latin script and analyzes it. It creates a **logical plan**, which represents the operations we want to perform. Then Pig converts the logical plan into a **physical execution plan**. The Hadoop processing framework can then execute these operations on the large dataset.

```
LOAD data
    ↓
FILTER marks > 60
    ↓
STORE result
```

One major advantage of Pig is that we describe what we want to do, while Pig handles much of the underlying processing complexity.

Apache Pig and Hadoop

Apache Pig is closely associated with the Hadoop ecosystem. Input data can be stored in HDFS (Hadoop Distributed File System). Pig can read that data, process it, and produce an output.

```
Apache Pig
     ↓
  Pig Latin
     ↓
Data Processing Plan
     ↓
Hadoop Processing
     ↓
   Result
```

Why Pig is Useful for Big Data

Processing a small dataset is relatively easy. But when the dataset contains 100 million or even 10 billion records, processing becomes much more complicated. Big-data technologies such as Hadoop are designed to process huge datasets across multiple machines. Apache Pig provides a simpler way to express the required data-processing operations.

Important Pig Operations

Operation	Purpose	Example
LOAD	Reads data.	data = LOAD 'students.csv';
FILTER	Selects records based on a condition.	passed = FILTER data BY marks > 60;
GROUP	Groups records based on a field.	grouped = GROUP students BY branch;
ORDER	Sorts data.	sorted = ORDER students BY marks DESC;

Operation	Purpose	Example
JOIN	Combines two datasets based on a common field.	joined = JOIN student BY Student_ID, marks BY Student_ID;
STORE	Saves the final result.	STORE result INTO 'output';

Pig Data Model

Pig can work with different data structures:

Atom: A single value, such as Asha, 85, or CSE.
Tuple: A collection of related fields, such as (Asha, CSE, 85). This is similar to a row.
Bag: A collection of tuples, similar to a collection of rows.
Map: A collection of key-value pairs, such as [name#Asha, branch#CSE, marks#85].

Pig vs MapReduce

MapReduce traditionally requires programming, often in Java. Apache Pig provides Pig Latin, which is more concise for many data-processing tasks. Therefore, Pig provides a higher-level abstraction over the underlying data-processing framework and makes many large-scale data-processing tasks easier to express.

Real-World Example: E-Commerce

Imagine an e-commerce company has millions of customer transactions and wants to find how many products were purchased in each category. Pig can load the transaction data, group it by category, calculate the required information, and store the result.

```
LOAD the transaction data
        ↓
GROUP by category
        ↓
Calculate required information
        ↓
STORE the result
```

Final Explanation to Remember

Apache Pig is a high-level platform for processing large datasets. It uses a scripting language called Pig Latin. We give Pig instructions such as LOAD, FILTER, GROUP, JOIN, and STORE, and Pig converts those instructions into an execution plan that can be processed using the Hadoop ecosystem.

Easy flow to remember: Large Data → Pig Latin → Data Transformation → Hadoop Processing → Result